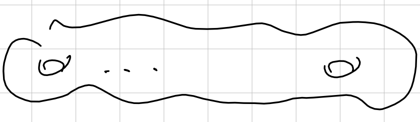
S^n, ~~$S^1 \times S^1$~~, $\mathbb{RP}^n$, 

$(n \geq 2)$

$\downarrow$

$\pi_1 = 0$

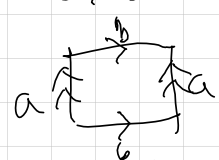
2-dim mfd

$L(p,q)$

$D^1 \times S^1 / e^{2\pi i p}, e^{2\pi i q}$

$M \# N$. $\wedge$ $\vee$

$[0,1]^2$



torus.

$\mathbb{Z} \times \mathbb{Z}$

klein.

klein. free product.

ababab$b$.

$\langle a, b \mid aba = b \rangle$.

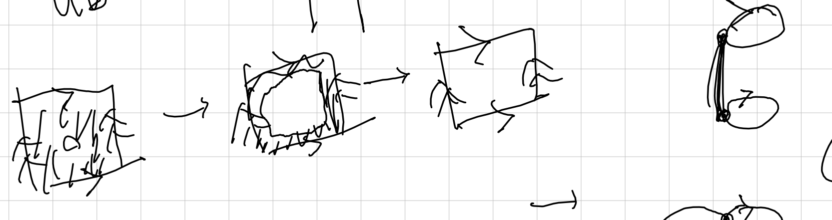
$aba^{-1}b^{-1} = e$.

$ab = ba \to$ group $+\mathbb{Z}$.

$\langle a, b \mid ab = ba \rangle$

$\mathbb{Z} \times \mathbb{Z}$

$S^1 \vee S^1$



$S^1 \vee S^2$

$T : \mathbb{Z} \times \mathbb{Z}$

$S^1 \vee S^1$

$\mathbb{Z} * \mathbb{Z}$

$T \# \cdot \mathbb{Z} * \mathbb{Z}$

$K \# \cdot \mathbb{Z} * \mathbb{Z}$

theorem

Theorem

$S^1 \vee S^1$

$\mathbb{Z} * \mathbb{Z}$

S^1

$\mathbb{Z}$

$S^3 : 0$.

$\mathbb{R}^3 : 0$.

M : n-dim with $n \geq 3$

then $\pi_1(M) \simeq \pi_1(M \setminus \cdot)$.

$\simeq \pi_1(M \setminus 2pt)$

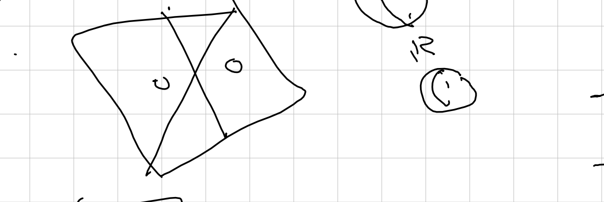
$\simeq \pi_1(M \setminus \{x_i\})$.

$\pi_1(S^n) = 0$

$n \geq 2$.

$\pi_1(\mathbb{RP}^n) = \mathbb{Z}_2$

$(n \geq 2)$.



tree

$S^1 \vee S^1 \vee \mathbb{R}$.

$\mathbb{Z} * \mathbb{Z}$

: $S^1 \vee S^1 \vee S^1$

$\mathbb{Z} * \mathbb{Z} * \mathbb{Z}$

Example.

$S^2 / N = S$

N

$\to \pi_1(\ldots)$

$S^1 \vee S^2$

S^1 : 0-cell 1개

1-cell 1개.

S^2 : 0-cell 1개

2-cell 1개

$S^1 \vee S^1$

$\mathbb{RP}^2$: 0-cell : 1개

1-cell : 1개

2-cell 1개

$aba^{-1}b^{-1} = 0$.

$\langle a, b \mid ab = ba \rangle$

$\mathbb{Z} * \mathbb{Z} / \alpha$

$\mathbb{RP}^2 : \mathbb{Z}_2$ $\langle a \mid a^2 \rangle$

$a^2 = 1$

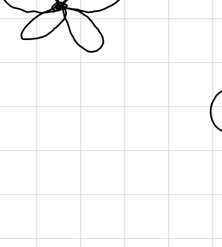
$\mathbb{RP}^3$: 0-cell 1개

1-cell 1개

2-cell 1개

3-cell 1개

$\pi_1(\mathbb{RP}^3) = \pi_1(\mathbb{RP}^2) = \mathbb{Z}_2$.



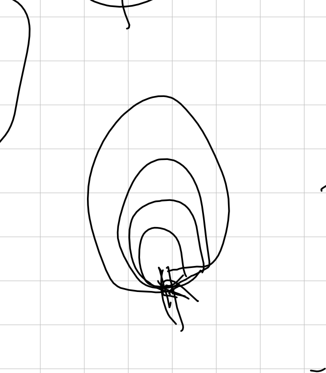
$x^2+y^2+z^2=1$

$\cup$ $x^2+y^2 \leq 1$

$\cup$ $(0,0,z)$ $-1 \leq z \leq 1$.

$(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$

$\mathbb{R}/\mathbb{Z}$



S^1

$z = x^4 + y^4$

$\downarrow$ exp

$p = 0$

$K_1 = f(4, 2)$

$K_2 = f(4, \ldots)$

$\frac{1}{r^2 + z}$.

$\int \frac{1}{z^2} dz$

$2\pi i$

$2\pi i \cdot (\ldots)$

0